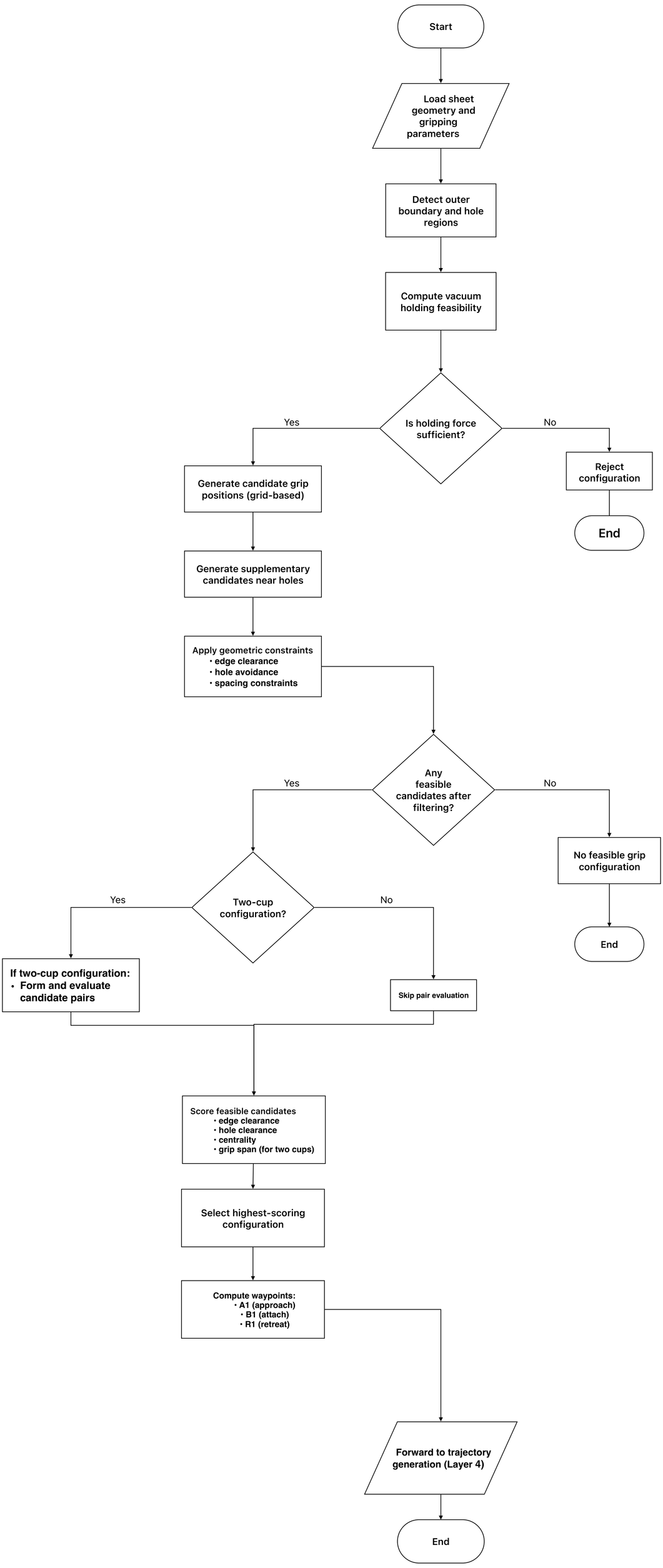
Start
Load sheet geometry and gripping parameters
Detect outer boundary and hole regions
Compute vacuum holding feasibility
Is holding force sufficient?
Yes
No
Reject configuration
End
Generate candidate grip positions (grid-based)
Generate supplementary candidates near holes
Apply geometric constraints
• edge clearance
• hole avoidance
• spacing constraints
Any feasible candidates after filtering?
Yes
No
No feasible grip configuration
End
Two-cup configuration?
Yes
No
If two-cup configuration:
• Form and evaluate candidate pairs
Skip pair evaluation
Score feasible candidates
• edge clearance
• hole clearance
• centrality
• grip span (for two cups)
Select highest-scoring configuration
Compute waypoints:
• A1 (approach)
• B1 (attach)
• R1 (retreat)
Forward to trajectory generation (Layer 4)
End